

A Gillespie Approach to an Acid-Base Indicator Equilibrium

Abstract

Traditionally, Acid-Base indicator equilibrium data would be modelled via deterministic ordinary differential equations (ODE), resulting in macroscopic and continuous predictions of a system's behaviour. However, by modelling this system via the Gillespie algorithm, we can obtain microscopic and discrete predictions that enable a full distribution of states to be observed.

The optimal concentration of bromophenol blue indicator was used (Absorbance=1) with acid, base, and buffer added to separate solutions and each analysed via UV-Vis. The data obtained was plotted and put through defined functions to extract peak absorbance, $\lambda_{\max}$, pKa (4.172), and ultimately probability of a molecule being deprotonated at equilibrium ($p=0.402$). Translating this new set of data into a defined function, a binomial distribution of deprotonated molecules was yielded, with the relative width of the distributions being inversely proportional to $\sqrt{N}$, and a clear convergence of distributions towards the mean ($p \times N$) with increasing N .

Theory & Code

The Acid-Base indicator equilibrium is given by $HIn^- \rightleftharpoons H^+ + In^{2-}$, where HIn^- is bromophenol blue's natural anionic form, and the rate constant of the forward and backward reaction can be given as k_f and k_b respectively.

The Gillespie algorithm states that the current state of the system (i.e. number of protonated molecules and number of deprotonated molecules) governs the propensities of the transitions to the next state. Since these propensities are dependent on only the current state and not the history of the system, it is Markovian, and thereby memoryless. Due to the total number of molecules being conserved, the pair of variables is dependent. Since the sum of these variables is fixed, the system has only one degree of freedom, and so only one variable may be tracked (deprotonated molecules in this case). We may naively classify the distribution of the total count of desired molecules in the system as a Bernoulli distribution since the molecules only have two options: protonated or deprotonated. However, while every molecule individually follows a Bernoulli distribution, the total number of desired molecules is the sum of independent Bernoulli random variables, and thus, is binomially distributed. This only holds provided the classes of molecules are independent of each other, and in this case, this is directly caused by the buffered $[H^+]$.

Three data sets of the UV-Vis spectra of acid, base, and buffer were passed through 'find_absorbance(data)' to obtain peak absorbance; 'find_wavelength(data)' to yield $\lambda_{\max}$; 'abs_at(data)' to return absorbance value at 589 nm which is where maximum sensitivity to $[In^-]$ concentration can be observed.

These obtained values were then passed through another set of functions: 'fraction(abs1, abs2, abs3)'; 'henderson_hasselbach(pH, f)'; 'Ka_calc(pKa)'; this returns the fraction of deprotonated molecules ($f=0.402$), the pKa of the indicator at equilibrium ($pKa=4.172$), and the dissociation constant of the equilibrium ($Ka=6.736 \times 10^{-5}$).

With all necessary values calculated, we can perform the Gillespie algorithm: 'Gillespie(N, t_max, forward_k, reverse_k)'. This function, at its root, is reliant on k_f (forward_k) and the product of k_b and $[H^+]$ • (reverse_k). These two parameters, along with current molecule count of both states, set the propensities of the two transitions, protonation or deprotonation, and the sum of said propensities gives us the overall propensity of any transition happening, be it protonation or deprotonation. In other words, the overall likelihood of a reaction happening is the sum of the likelihoods of each specific type of reactions. As mentioned before, Gillespie is a memoryless process, and so the time taken between each reaction can be given by τ which is a drawn value from an exponential distribution with mean equal to the reciprocal of the total reaction propensity. The function utilises the built-in function, 'np.random.exponential', to compute this. For us to decide which reaction occurs, a number is picked randomly from a uniform distribution over the range 0 to 1. This value is then compared to the fractional forward (deprotonation) propensity. If this value is below the threshold of the fractional propensity, deprotonated molecule count increases by one and

protonated decreases by one, and if the value is above the threshold, then vice versa. As each reaction occurs, time taken increases by one tick, because, as we've established before, τ is the time between each reaction. Ultimately, the function returns with two data sets: time, and deprotonated molecule count.

To plot an accurate distribution of deprotonated molecules at equilibrium, we must utilise the functions 'stationary_count_split(data, N, window, flat, repeat)' and 'stationary_count(N, t_max, forward_k, reverse_k, window, flat, repeat)'. The former of the two compares rolling means over adjacent windows, detecting where their rate of change flattens i.e. gives the cut off point where equilibrium is reached. The latter then discards the transient data leading up to the stationary counts of deprotonated molecules at equilibrium. After trimming the data, plotting of histograms faces the problem of the counts of molecules being discrete, and so these histograms require inter-aligned bins. The solution to this is the function 'binomial_plot(N)' generating an array with regular intervals, starting at -0.5 with intervals of 1, and ending at number+1.5, allowing the edges of the bins to align with integers.

With our ducks all in row, we may now explore, compare, and analyse the plots.

Key:

data = desired input data set(s) for functions

pH = pH at which equilibrium resides in

f = fraction of desired molecule (deprotonated)

N = total molecule count

t_max = maximum length of unit time for the simulation

forward_k = rate constant for forward reaction (k_f)

reverse_k = rate constant for reverse reaction ($[H^+] \cdot (\text{reverse_k})$)

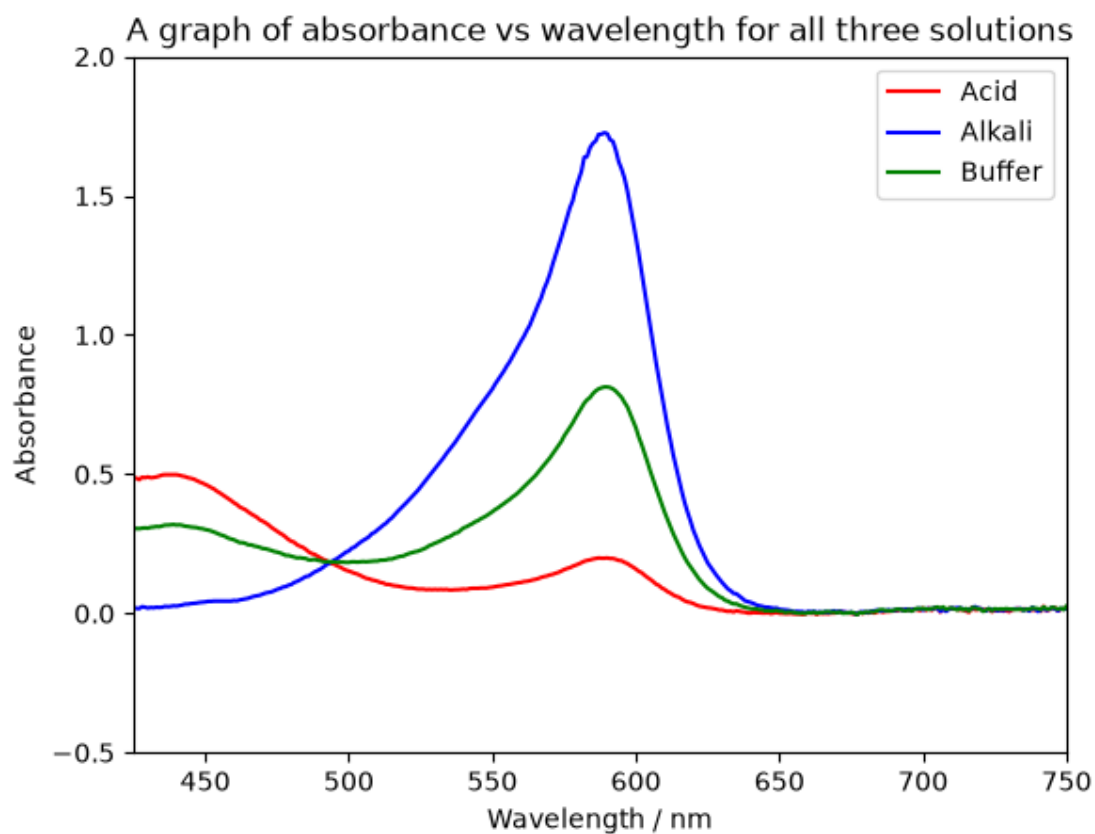
window = number of data points being averaged

flat = maximum rate of change to be accepted before data is declared "flat"

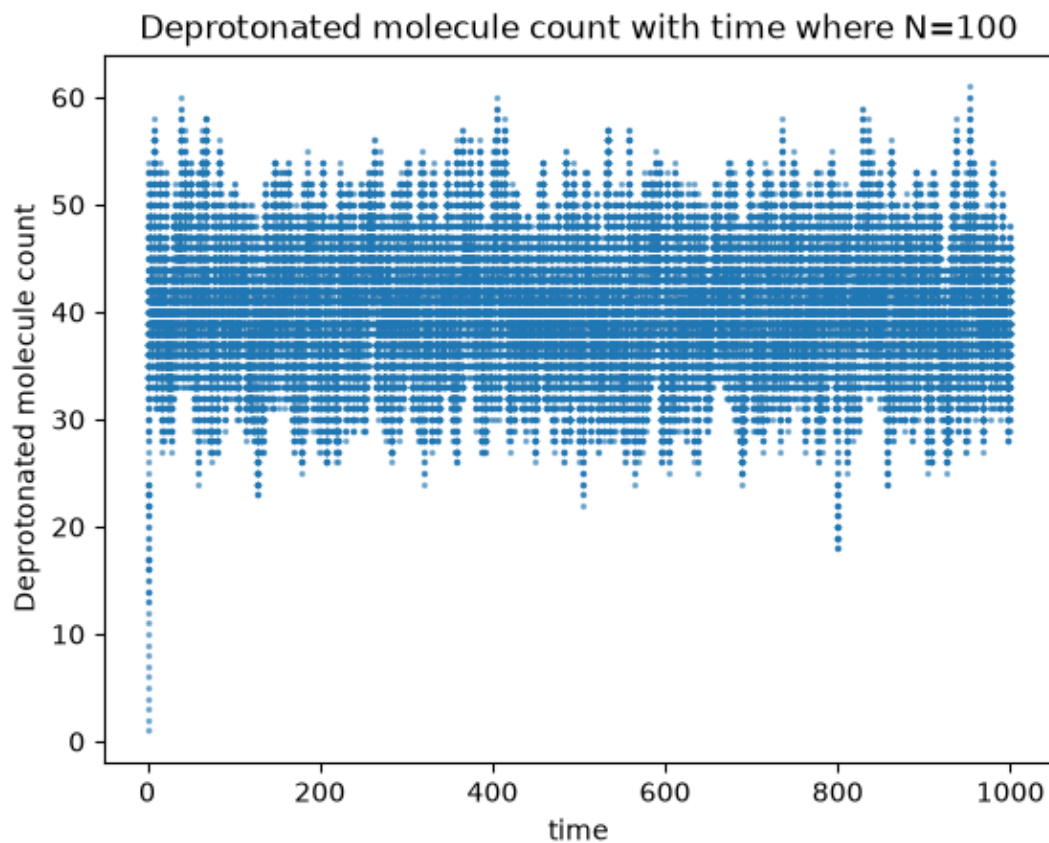
repeat = number of consecutive flat readings required for data to be accepted

Results

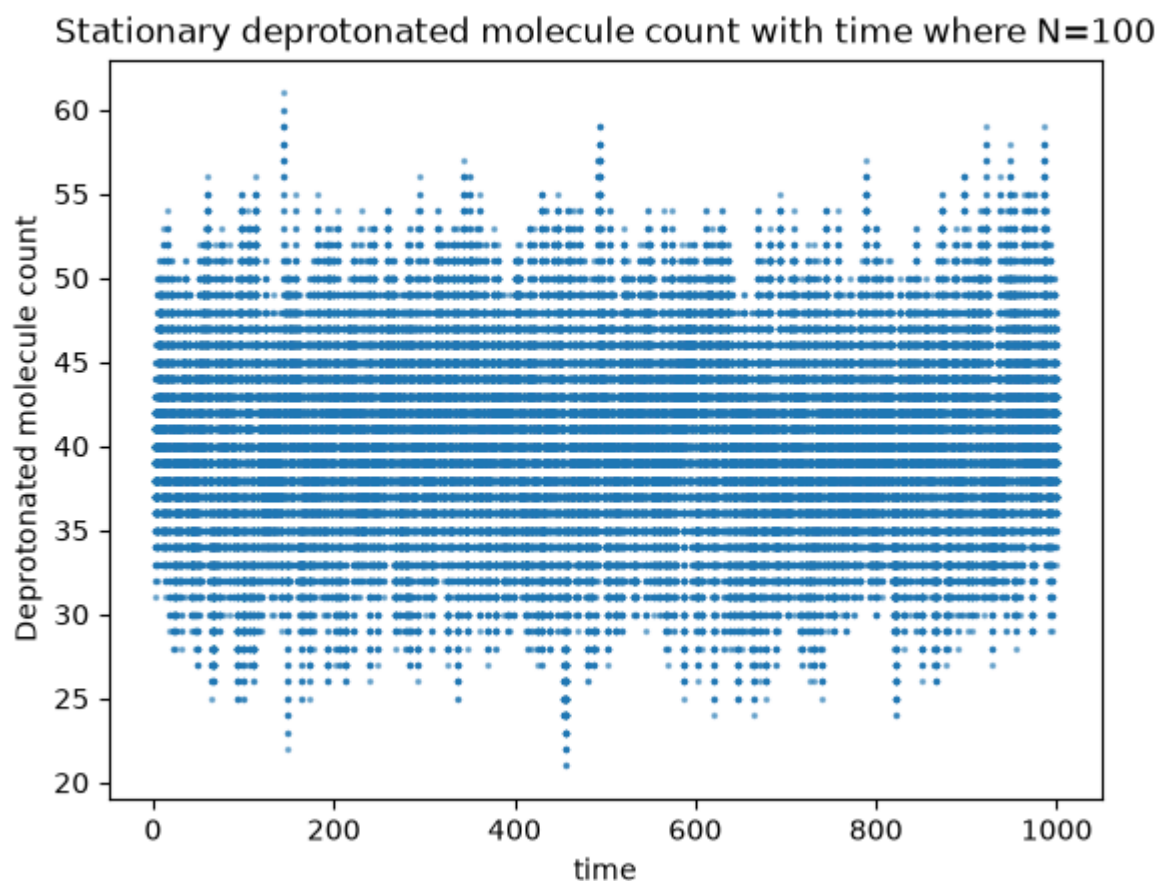
Absorbance Against Wavelength for Acid, Alkali, and Buffer Fig. 1



Deprotonated Molecule Count Over 1000 Units of Time – With Build Up Fig. 2

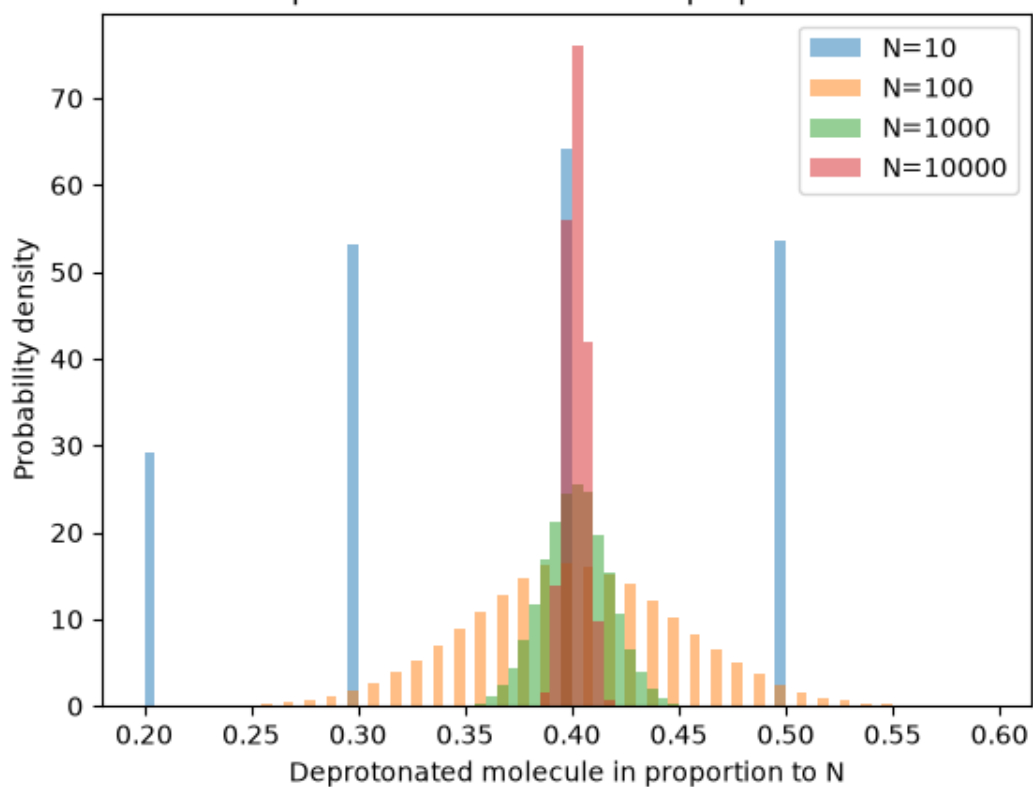


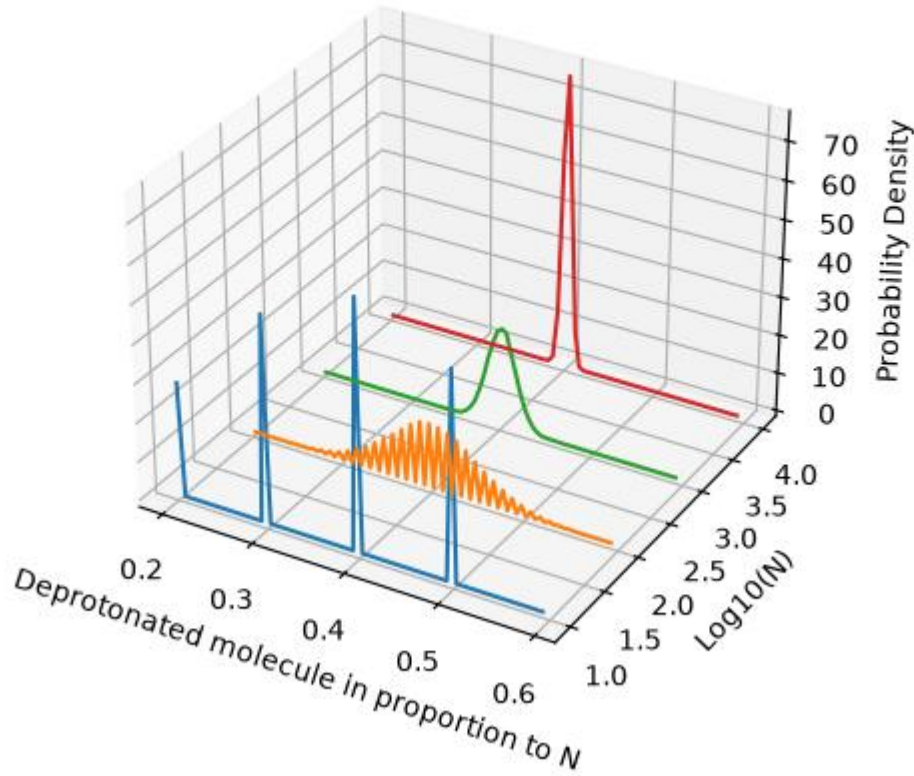
Deprotonated Molecule Count Over 1000 Units of Time – Stationary Counts Fig. 3



A Distribution of Deprotonated Molecules Across Multiple N Fig. 4

A distribution of deprotonated molecules in proportion to N at different N





Analysis

Our target molecules are deprotonated molecules, meaning the desired $\lambda_{\max}$ is 589 nm. It can be observed from the graph that alkali absorbance is far higher than the other two, acid and buffer. This is because the specific species In^{2-} which absorbs 589 nm light is at its greatest concentration in the alkali solution. As a qualitative confirmation, the solution was observed to be deep blue/violet in colour, giving off the complementary colour to the colour of light absorbed at 589 nm.

A comparison between Fig. 2 and Fig. 3 highlights the ‘transient’ counts. Since the simulation starts at $In^{2-} = 0$, the reaction is not at equilibrium and must climb up to it. For $N = 100$, the mean value would be 40.2, and this is confirmed by the density of counts (molecules) at the 40 values of the y-axis.

The collective plot demonstrates that as N increases, the distribution of deprotonated molecules concentrates towards p , the fraction of deprotonated molecules, demonstrating that relative width of the distributions being inversely proportional to $\sqrt{N}$, which we established previously. This is presented more clearly in Fig. 5 where we plot the distribution as a function of N , more specifically $\log_{10}(N)$.

Given $HIn^- \rightleftharpoons H^+ + In^{2-}$, forward rate constant = k_f , and backward rate constant = k_b .

$$\frac{d[HIn^-]}{dt} = k_b[In^{2-}][H^+] - k_f[HIn^-]$$

Equation. 1

Since concentration of all species remain constant at equilibrium, $\frac{d[HIn^-]}{dt} = 0$.

$$k_f[HIn^-] = k_b[In^{2-}][H^+]$$

Equation. 2

$$\frac{k_f}{k_b} = \frac{[In^{2-}][H^+]}{[HIn^-]}$$

Equation. 3

Since $\frac{k_f}{k_b} = Ka$, we can rearrange Equation. 3 to give:

$$\frac{Ka}{[H^+]} = \frac{[In^{2-}]}{[HIn^-]}$$

Equation. 4

We know that $Ka = 6.736 \times 10^{-5}$ and $[H^+] = 10^{-4}$, and so $\frac{[In^{2-}]}{[HIn^-]} = 0.674$, but what we are after is the proportion of deprotonated molecules $\frac{[In^{2-}]}{[HIn^-] + [In^{2-}]}$. We can accomplish this by finding the reciprocal of $\frac{[In^{2-}]}{[HIn^-]}$, adding 1 to give $\frac{[HIn^-]}{[In^{2-}] + [HIn^-]}$, then finding the new reciprocal, giving us $\frac{[In^{2-}]}{[HIn^-] + [In^{2-}]} = 0.402$, the probability of any one molecule being deprotonated at equilibrium.

It is at this point that we can see the distribution modelled by the Gillespie simulation narrowing towards the value obtained via deterministic ODE.

Author's Notes

The stochastic simulation methodology, embodied by the Gillespie algorithm here, appears throughout finance, acting as the foundation for many different classes of models. Since the Gillespie algorithm is memoryless with constant intensity, the next step to consider as this project's natural descendent would be to explore a history-dependent, self-exciting intensity i.e. the Hawkes Process where each event increases the likelihood of future events.